

The Erdős–Gyárfás conjecture for graphs of diameter at most three

Special Project*

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Abstract

Erdős and Gyárfás conjectured that every graph of minimum degree at least 3 contains a cycle whose length is a power of two. We prove that every graph, finite or infinite, with minimum degree at least 3 and diameter at most 3 contains a cycle of length 4 or 8. This extends the recent diameter-two result of Carr. The proof is computer-assisted. Assuming a counterexample, we grow a finite subgraph embedded in it, branching over every way a missing neighbour or a short path can attach. All 183,943 terminal branches contain a 4- or 8-cycle, the search tree has 3,169 internal nodes, and no configuration ever needs more than 15 vertices. A separately written checker verifies the certificate. We also give a short human proof of the bipartite case.

1 Introduction

In 1995 Erdős and Gyárfás conjectured that every graph with minimum degree at least 3 contains a cycle whose length is a power of 2 [1]. The conjecture is Problem 64 on erdosproblems.com, and it remains open. Known results include:

- graphs of sufficiently large average degree (Liu and Montgomery [2]);
- 3-connected cubic planar graphs (Heckman and Krakovski [3]);
- claw-free graphs;
- P_8 -, P_{10} - and P_{13} -free graphs [4, 5, 6], the last with computer aid;
- graphs of diameter 2, which always contain a C_4 or a C_8 (Carr [7]).

Computer searches show that a counterexample has at least 24 vertices, and that none exists among graphs on at most 40 vertices or cubic graphs on at most 48 [8, 9]. Markström found cubic graphs on 24 vertices whose only power-of-two cycle length is 16 [10].

Theorem 1. *Every graph G (finite or infinite) with minimum degree at least 3 and diameter at most 3 contains a cycle of length 4 or 8.*

Corollary 2. *The Erdős–Gyárfás conjecture holds for all graphs of diameter at most 3.*

Two remarks put the statement in context.

*Produced with Claude (Anthropic) in an interactive session. The computer search and the independent checker are included with this note.

- **Infinite graphs.** The infinite 3-regular tree has minimum degree 3 and no cycles at all, so the conjecture fails for infinite graphs in general. Theorem 1 shows that bounded diameter (at most 3) restores it.
- **Both lengths are needed.** K_4 has no 8-cycle, and the graph found in Section 3 has diameter 3 and no 4-cycle.

2 The method

Fix integers D (the diameter bound), δ (the minimum-degree bound) and a set F of forbidden cycle lengths. In this note $D = 3$, $\delta = 3$ and $F = \{4, 8\}$. A *counterexample* is a graph G with minimum degree at least δ , diameter at most D , and no cycle whose length lies in F .

A *configuration* is a finite graph H on vertices $\{0, \dots, n-1\}$. An *embedding* of H into G is an injective map $\varphi : V(H) \rightarrow V(G)$ that sends edges to edges. A *requirement* of H is either of the following:

- a vertex v with $\deg_H(v) < \delta$;
- a pair $\{a, b\}$ with $\text{dist}_H(a, b) > D$, where the distance is infinite if a and b lie in different components.

Witnesses. Let $n = |V(H)|$, and call n and $n+1$ *fresh*.

- For a degree requirement v , the witnesses are the single edges vu , where either u is a vertex of H that is distinct from v and not adjacent to it, or $u = n$ is fresh.
- For a distance requirement $\{a, b\}$, the witnesses are the paths $a = w_0, w_1, \dots, w_k = b$ with $1 \leq k \leq D$ and distinct vertices. Each interior vertex is either a vertex of H other than a and b , or fresh. Fresh vertices are numbered $n, n+1, \dots$ in order of appearance, which removes relabelling symmetry.

Adding a witness to H gives a configuration $H + w$.

Proof trees. A proof tree for H selects one requirement ρ of H and, for every witness w of ρ , records one of two outcomes:

- $H + w$ contains a cycle whose length lies in F (a *dead branch*); or
- a proof tree for $H + w$ is given.

Note that if H has no requirement, no proof tree for H exists. In that case H itself is a finite counterexample.

Lemma 3 (Soundness). *Suppose a finite proof tree exists for the one-vertex configuration. Then no counterexample exists.*

Proof. Suppose G is a counterexample. We maintain a configuration H that has a proof tree, together with an embedding $\varphi : H \rightarrow G$. Start with $H = K_1$, mapping its vertex to any vertex of G .

Let ρ be the requirement selected at H . We find a witness w of ρ and extend φ to an embedding of $H + w$, as follows.

- *Degree requirement v .* Since $\deg_G \varphi(v) \geq \delta > \deg_H(v)$, the vertex $\varphi(v)$ has a neighbour $x \notin \varphi(N_H(v))$. If $x = \varphi(u)$ for some $u \in V(H)$, then $u \neq v$ and u is not adjacent to v in H , so $w = vu$ is a witness. Otherwise $w = vn$ with n fresh, and we set $\varphi(n) = x$.
- *Distance requirement $\{a, b\}$.* Let P be a shortest $\varphi(a)$ – $\varphi(b)$ path in G . It has length $k \leq D$, and its vertices are distinct. Pull P back to H : an interior vertex of the form $\varphi(u)$ becomes u , and every other interior vertex becomes a fresh vertex, numbered in order. The result is a witness w . Define φ on the fresh vertices accordingly.

In both cases the extended map is injective. Fresh vertices go outside $\varphi(V(H))$, and the vertices of P are distinct. It also sends edges to edges, so it is an embedding of $H + w$.

The proof tree assigns an outcome to w . If the branch is dead, $H + w$ contains a cycle with length in F , and its image under the embedding is such a cycle in G , a contradiction. Otherwise we continue with the proof tree of $H + w$. The tree is finite, so a dead branch is reached after finitely many steps. $\square$

The argument never uses finiteness of G : degrees in G may be infinite, and only finitely many vertices are ever inspected. So Theorem 1 follows from Lemma 3 once a proof tree for $D = 3$, $\delta = 3$, $F = \{4, 8\}$ has been exhibited. That is what the computation does.

3 Computation

The search program `search.py` builds proof trees depth-first. At each configuration it considers every requirement, preferring distance requirements, and selects the one with the fewest witnesses that are not immediately dead. A branch is dead when one of the newly added edges lies on a cycle whose length is in F . Running the program with $D = 3$, $\delta = 3$, $F = \{4, 8\}$ produces a complete proof tree, saved as `proof_tree.json`.

Instance	internal nodes	dead branches	max. $ V(H) $	time
$D = 2$, $F = \{4, 8\}$ (Carr’s theorem)	38	180	8	< 1 s
$D = 3$, $F = \{4, 8\}$ (Theorem 1)	3,169	183,943	15	6 s

Independent verification. The checker `checker.py` was written separately and shares no code with the search. It replays the tree from K_1 and, at every node, does three things.

1. It confirms, using its own breadth-first search, that the recorded requirement really is a requirement of the current configuration.
2. It regenerates the full set of witnesses itself and confirms that each one is covered by a branch of the tree. Edges already present in H are ignored when matching.
3. For every dead branch, it confirms with an independent cycle enumerator (the `simple_cycles` routine of `networkx`, with a length bound) that the configuration contains a 4- or 8-cycle.

The checker accepts the certificate in about 35 seconds.

Controls. The same program behaves correctly on instances whose answers are known.

- With $D = 2$ it reproduces Carr's theorem.
- With $F = \{4\}$ it returns a genuine 10-vertex graph with minimum degree 3, diameter 3 and no 4-cycle. That graph necessarily contains 8-cycles.
- With $\delta = 2$ it returns the triangle, a valid counterexample to the weakened statement.
- With $D = 4$ and $F = \{4, 8\}$ it does not terminate within our time limit. This is consistent with the expectation that Markström-type graphs, which have no 4- or 8-cycle, exist at diameter 4.

4 A human proof of the bipartite case

The computer proof is a large case analysis. To illustrate the kind of argument it contains, here is a short proof of one special case.

Proposition 4. *Every bipartite graph with minimum degree at least 3 and diameter at most 3 contains a cycle of length 4 or 8.*

Proof. Suppose G has no C_4 . Being bipartite, G then has girth at least 6.

Every path $abcde$ with four edges lies on a 6-cycle. The vertices a and e lie on the same side of the bipartition, so $\text{dist}(a, e)$ is even. It is at most 3, so it equals 2. Let m be a common neighbour of a and e . We check that m is not on the path:

- $m = b$ would make $bcde$ a 4-cycle;
- $m = d$ would make $abcd$ a 4-cycle;
- $m = c$ is impossible, because c lies on the same side as a .

Hence $abcde m$ is a 6-cycle.

Constructing the 8-cycle. A path with four edges exists because the minimum degree is 3, so by the claim there is a 6-cycle $c_0 c_1 \cdots c_5$. It has no chord, since a chord would create a C_4 or join two vertices on the same side. So c_1 has a neighbour z outside the cycle, and z has two neighbours $y_1, y_2 \neq c_1$. Neither y_j lies on the cycle: it would be c_3 or c_5 , creating the 4-cycle $c_1 z c_3 c_2$ or $c_1 z c_5 c_0$.

Fix $j \in \{1, 2\}$. By the claim, the path $y_j z c_1 c_2 c_3$ closes into a 6-cycle through a common neighbour m_j of y_j and c_3 , with $m_j \notin \{z, c_1, c_2\}$. Suppose $m_j \notin \{c_4\}$. Then

$$y_j z c_1 c_0 c_5 c_4 c_3 m_j$$

is an 8-cycle, since its eight vertices are distinct. Finally, $m_1 = m_2 = c_4$ is impossible: it would give two common neighbours y_1, y_2 of z and c_4 , that is, a 4-cycle. $\square$

When the graph is not bipartite, the triangles, 5-cycles and 7-cycles allowed by C_4 -freeness multiply the cases. This is where the computer search takes over.

5 Discussion

Theorem 1 is sharp in the sense that the method stops working at diameter 4, where 4-cycles and 8-cycles alone are not enough. Markström’s cubic graphs on 24 vertices, whose only power-of-two cycle length is 16, necessarily have diameter at least 4, because a cubic graph of diameter 3 has at most 22 vertices. Two questions are natural.

1. Does every graph with minimum degree at least 3 and diameter at most 4 contain a cycle of length 4, 8 or 16? This is the next case of the conjecture. The same search applies, with $F = \{4, 8, 16\}$.
2. Is there a short human proof of Theorem 1? The proof tree is small enough, at under 3,200 internal nodes, that a structured human argument may well exist.

Files. The files `search.py`, `checker.py` and `proof_tree.json.gz` (394 KB) are in the same directory as this note.

- Reproduce the proof tree with `python3 search.py 3 4,8 3 18 -tree proof_tree.json`.
- Check it with `python3 checker.py proof_tree.json`.

References

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